

Q. Let's consider a dataset of weather condition & whether or not a person played golf. The features are

- outlook : sunny, overcast, rainy
- Temperature : Hot, mild, cool
- Humidity : High, Normal
- windy : True, false

The target variable is "play golf" (yes/no)

outlook	Temp	Humidity	windy	play golf
sunny	Hot	High	False	No
sunny	Hot	High	true	No
overcast	Hot	High	False	Yes
Rainy	mild	High	False	Yes
Rainy	cool	Normal	False	Yes
Rainy	cool	Normal	true	No
overcast	cool	Normal	true	Yes
sunny	mild	High	False	No
sunny	cool	Normal	False	Yes
Rainy	mild	Normal	False	Yes
sunny	mild	Normal	true	Yes
overcast	mild	High	true	Yes
overcast	Hot	Normal	False	Yes
Rainy	mild	High	true	No

1) Use the ID3 algo to construct a decision tree for this dataset using pen & paper

calculate the entropy of information gain for each feature at each step.

3) Show the final decision tree with its decision rules.

4) Analyze the tree & discuss its implications for predicting whether a person will play golf based on weather conditions.

→ In a given dataset containing weather condition & target variable "play golf"

The attributes are

- 1) outlook (sunny, overcast, rainy)
- 2) Temperature
- 3) Humidity
- 4) windy

calculating Entropy

Total records = 14

Yes = 9

No = 5

$$\text{Entropy}(S) = -P_{\text{Yes}} \log_2(P_{\text{Yes}}) - P_{\text{No}} \log_2(P_{\text{No}})$$

$$= -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14}$$

$$= -(0.643)(-0.64) - (0.357)(-1.48)$$

$$= 0.940$$

$$\therefore \text{Initial Entropy} = 0.94$$

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Information gain for each feature

1. outlook

sunny (5 records)

Yes = 2

No = 3

$$\text{Entropy (sunny)} = -\frac{2}{5} \log_2 \left(\frac{2}{5} \right) - \frac{3}{5} \log_2 \left(\frac{3}{5} \right)$$

$$= 0.971$$

similarly

overcast (4 records)

Yes = 4, No = 0

Entropy = 0

Rainy (5 records)

Yes = 3, No = 2

Entropy = 0.97

weighted entropy

$$= \frac{5}{14} (0.971) + \frac{4}{14} (0) + \frac{5}{14} (0.971)$$

$$= 0.693$$

$$\text{Gain(outlook)} = 0.94 - 0.693 = 0.247$$

2. Temperature

Hot (4 records)

Yes = 2, No = 2

$$\text{Entropy} = -0.5 \log_2 (0.5) - 0.5 \log_2 (0.5) \\ = 1$$

mild (6 records)

Yes = 4, No = 2

$$\text{Entropy} = -\frac{4}{6} \log_2 \frac{4}{6} - \frac{2}{6} \log_2 \frac{2}{6} \\ = 0.918$$

cool (4 records)

Yes = 3, No = 1

$$\text{Entropy} = -\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4} \\ = 0.811$$

weighted entropy

$$= \frac{4}{14} (1) + \frac{6}{14} (0.918) + \frac{4}{14} (0.811)$$

$$= 0.911$$

Information gain

$$\text{Gain}(\text{temperature}) = 0.940 - 0.911$$

$$= 0.029$$

Information gain for humidity

High (7 records)

Yes = 3, No = 4

$$\text{Entropy} = -\frac{3}{7} \log_2 \frac{3}{7} - \frac{4}{7} \log_2 \frac{4}{7}$$

$$= 0.985$$

Normal (7 records)

Yes = 6, No = 1

$$\text{Entropy} = -\frac{6}{7} \log_2 \left(\frac{6}{7}\right) - \frac{1}{7} \log_2 \left(\frac{1}{7}\right)$$

$$= 0.591$$

weighted entropy

$$= \frac{7}{14} (0.985) + \frac{7}{14} (0.591)$$

$$= 0.787$$

$$\text{Gain}(\text{Humidity}) = 0.940 - 0.787$$

$$= 0.153$$

Information gain for windy

True (6 records)

Yes = 3

No = 3

$$\text{Entropy} = 1$$

false (8 records)

yes = 6, No = 2

Entropy = 0.811

weighted entropy

$$= \frac{6}{14} (1) + \frac{8}{14} (0.811)$$

$$= 0.892$$

$$\text{Gain (windy)} = 0.940 - 0.892$$

$$= 0.048$$

final comparison

Attribute	Information gain
outlook	0.247
Humidity	0.153
windy	0.048
temperature	0.029

$\therefore$ Highest gain = outlook

